# Testing Human Debiasing Prompts in LLMs: Prompt-Level Evidence from 22,773 Anchoring Bias Trials

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**Abstract**

Do human debiasing techniques work on LLMs? We test one prompt implementation per technique for five approaches from the cognitive bias literature—Devil's Advocate, SACD, Premortem, Outside View, and a Random Control—across two studies totaling 22,773 anchoring bias trials (14,152 main study + 8,621 multi-domain extension).

**For these prompts: not reliably.** Some techniques help (SACD achieves 93.7% of unanchored baseline, though confounded by its 6-turn structure vs. 3 turns for others), but one makes things worse: our specific Devil's Advocate prompt produces responses at only 63.6% of baseline—*further from correct than doing nothing* (72.9%). Worse, this prompt looks best under the standard metric (susceptibility), which measures consistency rather than accuracy. Alternative prompts may perform differently; we test one implementation per technique.

Effectiveness varies dramatically: SACD shows the highest model variance, ranging from 48% to 128% of baseline. Point-estimate rankings vary across domains (CIs overlap). No technique consistently outperforms.

**Methodological recommendation:** We advocate reporting **MAD** (Mean Absolute Deviation from baseline) alongside aggregate metrics. Aggregate % of baseline can mislead: SACD's 93.7% masks 18.1% MAD because over/undershoots cancel.

**For practitioners:** Don't assume human techniques transfer. Test per-model, per-task. Report MAD alongside aggregates.

## 1 Introduction

Human debiasing techniques—Devil's Advocate, Premortem, deliberative reasoning—are increasingly applied to LLMs. But do they actually work? We test one prompt per technique across 22,773 trials and find transfer is not straightforward: SACD effectiveness ranges from 48% to 128% of baseline across models. Different prompt formulations may yield different results. Most concerning, standard evaluation metrics can recommend the wrong technique.

### 1.1 Two Metrics, Two Rankings

Prior work evaluates debiasing using **susceptibility**—the gap between high-anchor and low-anchor responses:

$$\text{Susceptibility} = |\bar{R}_{high} - \bar{R}_{low}| \tag{1}$$

Lower gap = less susceptible. But susceptibility measures *consistency*, not *accuracy*. A technique can achieve low susceptibility while landing far from the unanchored judgment.

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**Our approach:** We add **% of baseline**—proximity to the model's unanchored judgment:

$$\% \text{ of Baseline} = \frac{R_{technique}}{R_{baseline}} \times 100\% \tag{2}$$

100% = the model's anchor-free response (which may itself be incorrect).

### 1.2 Aggregate Metrics Hide Per-Trial Error

Even % of baseline can mislead when averaged (bidirectional errors cancel). SACD achieves 93.7% aggregate but 18.1% MAD. We report both metrics: % of baseline for comparability with prior work, MAD for per-trial accuracy (see Section 3.1 for formal definition). *Turn-count caveat:* SACD uses ∼6 turns vs. 3 for others; Random Control's strong performance (78.3%) suggests turn count itself matters.

### 1.3 Contributions

(1) **Aggregate metrics mask bidirectional error.** We demonstrate that aggregate % of baseline can mislead: SACD achieves 93.7% but 18.1% MAD, because over/undershoots cancel. We recommend reporting MAD alongside aggregates. (2) **Main study:** Deep validation (14,152 judicial trials, 10 models) confirming substantial model-specific variance (SACD: 48%–128%). (3) **Multi-domain extension:** Cross-domain evaluation (8,621 trials, 6 domains, 4 models) showing technique rankings vary by domain and metric. Combined total: 22,773 trials.

## 2 Related Work

### 2.1 Anchoring Bias in Human Judgment

Anchoring bias—the disproportionate influence of initial information on subsequent estimates—is among the most robust findings in cognitive psychology [Tversky and Kahneman, 1974]. Even experts are susceptible: Englich et al. [2006] demonstrated that experienced judges' sentencing decisions were influenced by random numbers generated by dice rolls. Effect sizes of $d = 0.6$–$1.2$ persist regardless of anchor source or participant awareness. Our experimental paradigm adapts this judicial sentencing design.

### 2.2 Cognitive Biases in LLMs

LLMs exhibit human-like cognitive biases [Binz and Schulz, 2023, Jones and Steinhardt, 2022, Chen et al., 2025], including anchoring [Huang et al., 2025, Song et al., 2026]. Unlike humans, LLMs enable exhaustive systematic testing.

### 2.3 Debiasing Techniques

We evaluate five techniques: **SACD** (iterative self-correction [Lyu et al., 2025]), **Devil's Advocate** (argue against initial response), **Premortem** (imagine decision failed [Klein, 2007]), **Random Control** (neutral filler to isolate turn-count effects), and **Outside View** (reference class forecasting [Sibony, 2019], excluded from rankings due to confound—see Section 6.3). Recent work on framing debiasing [Lim et al., 2026] shares conceptual overlap.

### 2.4 Evaluation Methodology

Standard anchoring evaluation uses **susceptibility** (high–low spread). We add **% of baseline** (proximity to unanchored judgment). Unlike the classic Anchoring Index [Jacowitz and Kahneman, 1995], which uses baseline in both numerator and denominator, our metric avoids circularity. Low AI with poor baseline proximity indicates consistent but wrong responses—the failure mode we observe with our DA implementation. We use free-form numeric responses rather than MCQs to avoid position and label biases documented in benchmark evaluations [**?**].

## 3 Methodology

### 3.1 Evaluation Metrics

We compare susceptibility (standard) with % of baseline (Section 1.1). Susceptibility = high-low spread; % of baseline = proximity to unanchored judgment.

**Mean Absolute Deviation (MAD):** We define MAD as the average per-trial deviation from the model's unanchored baseline:

$$\text{MAD} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{r_i}{b_m} - 1\right| \times 100\% \tag{3}$$

where $r_i$ is the response for trial $i$, $b_m$ is model $m$'s unanchored baseline response, and the sum is over all anchored trials. Lower MAD indicates responses closer to unanchored judgment. Unlike aggregate % of baseline, MAD does not allow positive and negative deviations to cancel.

**Why both?** Metrics can give divergent rankings—reducing spread may move responses away from baseline (Section 4.2).

**Baseline as reference, not ground truth.** Following Jacowitz and Kahneman [1995], we measure deviation from each model's unanchored baseline rather than an external ground truth. This operationalizes debiasing as *consistency*: a debiased model produces similar outputs regardless of irrelevant anchors. We make no claim that baselines are normatively correct—indeed, models disagree substantially (Table 2). The goal is to measure whether debiasing techniques restore anchor-independent reasoning, not whether that reasoning reaches any particular conclusion. This approach follows Kahneman & Tversky's original anchoring paradigm: baseline responses represent what the model would output absent anchoring influence, making deviation from baseline a direct measure of anchor effect. Alternative approaches (e.g., expert ground truth, legal guidelines) would conflate model capability with debiasing effectiveness.

**On proportional anchor design.** We set anchors proportionally to each model's baseline (±50%) to ensure comparable anchoring pressure across models with different baseline judgments. This is not circular: baselines are measured independently in anchor-absent conditions, then used to compute anchor values. A technique causing complete anchor-ignorance would score 100%—matching the model's unanchored judgment. The goal is to eliminate the influence of task-irrelevant numeric information on judgment.

### 3.2 Experimental Design

#### 3.2.1 Models

We evaluated 10 models across 4 providers:

| Provider | Models |
|---|---|
| Anthropic | Claude Haiku 4.5, Sonnet 4.6, Opus 4.6 |
| OpenAI | GPT-4.1, GPT-5.2, o3, o4-mini |
| DeepSeek | DeepSeek-v3.2 |
| Others | Kimi-k2.5 (Moonshot), GLM-5 (Zhipu) |

#### 3.2.2 Conditions

1. **Baseline**: Sentencing prompt with no anchor
2. **Low anchor**: Prosecutor demand at baseline $\times$ 0.5
3. **High anchor**: Prosecutor demand at baseline $\times$ 1.5
4. **Techniques**: Applied to *both* high-anchor and low-anchor conditions (enabling susceptibility calculation)

#### 3.2.3 Techniques Evaluated

| Technique | Description |
|---|---|
| Outside View | "What typically happens in similar cases?" (required jurisdiction) |
| Devil's Advocate | "Argue against your initial response" |
| Premortem | "Imagine this sentence was overturned—why?" |
| Random Control | Extra conversation turns with neutral content |
| SACD | Iterative self-administered cognitive debiasing |

#### 3.2.4 Temperature Conditions

Tested at t=0, 0.7, 1.0. No significant temperature$\times$technique interaction ($F(6, 8944) = 1.42$, $p = 0.203$); temperature effects <3pp. Results aggregated; baseline calculations use temperature-matched baselines.

#### 3.2.5 Trial Counts and Procedure

- **Total**: 14,152 judicial + 8,621 multi-domain = 22,773 analyzed
- **Per cell**: 30–90 trials (min $n = 30$ pre-specified; added trials for high-variance conditions)
- **Baselines**: 909 trials ( 90 per model)
- **Extraction**: Regex for integer months; 99.9% success
- **Anchors**: Proportional (high = 1.5$\times$baseline, low = 0.5$\times$)

Table 1: Trial distribution. Total unique trials: 14,152. Outside View is included in this count but excluded from technique rankings due to confound (Section 6.3). Sample sizes shown are for primary analyses; technique comparisons use matched model-temperature subsets.

| Condition | $n$ (analysis) |
|---|---|
| *Debiasing Techniques* | |
| SACD | 2,389 |
| Outside View | 2,423 |
| Random Control | 2,215 |
| Premortem | 2,186 |
| Devil's Advocate | 2,166 |
| *Control Conditions* | |
| Anchored (no technique) | 1,864 |
| Baseline (no anchor) | 909 |

#### 3.2.6 Statistical Analysis

Welch's t-test with Bonferroni correction (6 comparisons, $\alpha = 0.0083$) for hypothesis testing; percentile bootstrap (10,000 resamples, stratified by model) for CIs. Cohen's $d$ for effect sizes. Aggregates are trial-weighted means; see Table 5 for model-level. Trials collected via OpenRouter API (February 2026); full details in repository.

**Power analysis:** Model clustering (ICC=0.17) reduces effective $n$ to ≈60–70 per technique; powered to detect $d \geq 0.50$. Large effects ($d = 1.06$ SACD vs. DA) are reliably detectable; SACD–Premortem ($d = 0.08$) is underpowered.

### 3.3 Confounds and Limitations

**Edge models:** Opus 4.6 shows zero baseline variance (always 18mo); Haiku 4.5 has 85%+ refusal rates (surviving trials may exhibit selection bias). **Sensitivity analysis:** Excluding Opus shifts means <3pp; excluding Haiku shifts means <2pp. Rankings preserved in both cases: SACD remains #1 (91.2% excl. Haiku), Premortem #2, DA #4.

**Outside View:** Required jurisdiction specification, introducing potential secondary anchor. Excluded from rankings (Section 6.3).

## 4 Results

### 4.1 Baseline Responses

Unanchored baseline responses varied substantially across models:

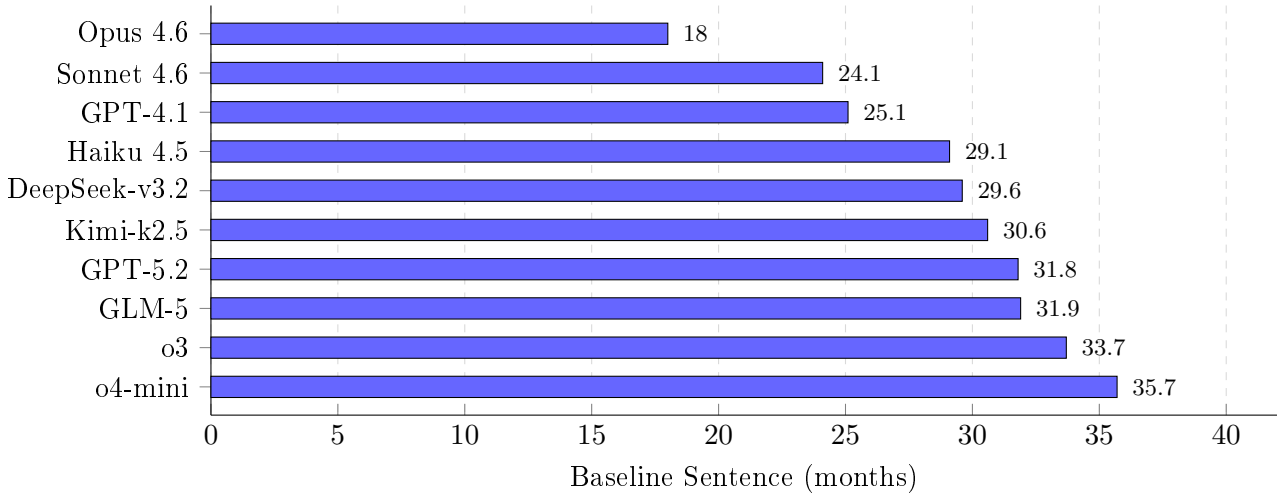


Figure 1: Model baseline variation. Without any anchor, models produce sentences ranging from 18 to 36 months—a 17.7-month spread. This variation motivates per-model anchor calibration.

| Model | Baseline Mean | SD |
|---|---|---|
| o4-mini | 35.7mo | 4.7 |
| o3 | 33.7mo | 5.6 |
| GLM-5 | 31.9mo | 5.7 |
| GPT-5.2 | 31.8mo | 5.7 |
| Kimi-k2.5 | 30.6mo | 7.4 |
| DeepSeek-v3.2 | 29.6mo | 8.0 |
| Haiku 4.5 | 29.1mo | 11.2 |
| GPT-4.1 | 25.1mo | 3.4 |
| Sonnet 4.6 | 24.1mo | 1.3 |
| Opus 4.6 | 18.0mo | 0.0 |

Table 2: Model baselines range from 18.0mo (Opus) to 35.7mo (o4-mini). Opus 4.6 shows zero variance; retained because rankings are robust to exclusion (see Limitations).

**Opus anomaly:** Opus 4.6 shows zero baseline variance (SD=0.0), consistently responding with exactly 18 months regardless of temperature. This may reflect deterministic reasoning or strong training priors for judicial contexts. We retain Opus because: (1) it represents a legitimate deployment scenario; (2) excluding post-hoc would inflate technique effectiveness. Sensitivity analysis (Limitations) shows rankings are robust to Opus exclusion.

### 4.2 Metric Divergence: Susceptibility vs. Baseline Proximity

Our core empirical finding: susceptibility and % of baseline give **divergent technique rankings**. *Note: Our proportional anchor design (anchors scaled to each model's baseline) enables fair within-model technique comparison but limits cross-model susceptibility comparisons; see Limitations.*

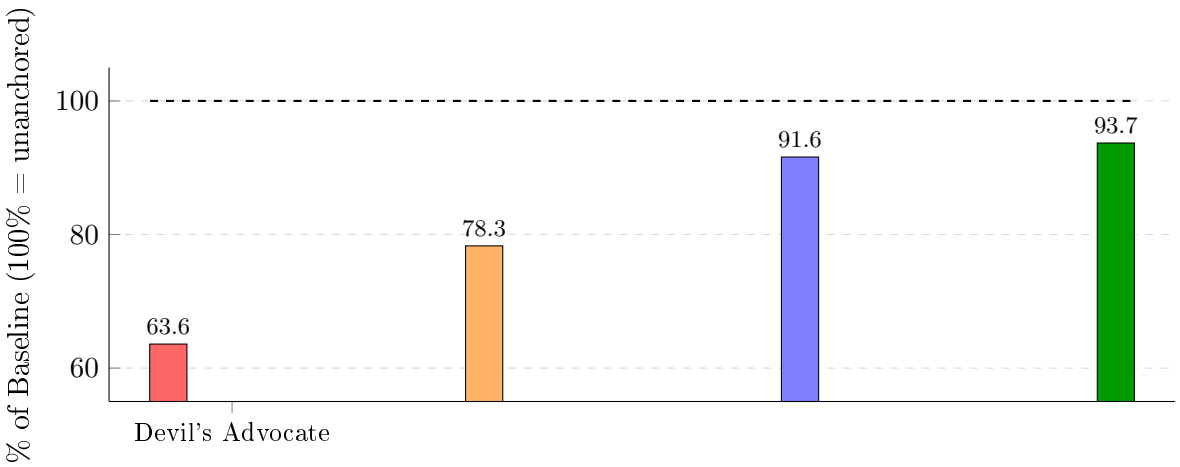


Figure 2: Technique responses as % of baseline. Dashed line = 100% (unanchored judgment). Our DA implementation keeps responses at 63.6% of baseline—consistently far from the unanchored judgment despite appearing "best" under susceptibility. SACD achieves 93.7%—closest to the model's unanchored judgment. *Note:* SACD's ranking is confounded by turn count (∼6 vs 3 turns); results reflect specific prompt implementations.

Table 3: Susceptibility vs. % of Baseline: Rankings diverge. Our Devil's Advocate implementation reduces susceptibility (−8.8%) yet performs worst on % of baseline (63.6%)—it achieves consistency by moving responses away from unanchored judgment. 95% CIs from bootstrap. Results reflect specific prompt implementations (Appendix).

| Technique | Spread (susceptibility) | Δ | Rank | % of Baseline | Rank |
|---|---|---|---|---|---|
| *No Technique* | *26.0pp* | *ref* | — | *72.9%* | *ref* |
| Devil's Advocate | 23.7pp | −8.8% | #1 | 63.6% [62, 65] | #4 |
| Random Control | 30.1pp | +15.8% | #2 | 78.3% [77, 80] | #3 |
| SACD | 36.3pp | +39.6% | #3 | 93.7% [92, 95] | #1 |
| Premortem | 45.2pp | +73.8% | #4 | 91.6% [90, 93] | #2 |

**Why the divergence?** Our DA implementation produces *consistent* responses (low spread) that remain *far from baseline*. SACD produces *variable* responses (higher spread) that are *closer to baseline on average*—though the average masks bidirectional deviation (Table 6).

**Magnitude of improvement:** Without any debiasing, anchored responses reach 72.9% of baseline—a deficit of 27.1 percentage points from the unanchored judgment. SACD achieves 93.7%, reducing this deficit to 6.3pp. This represents a reduction of 20.8pp, though the residual deficit and bidirectional variance (MAD=18.1%) remain limitations.

**Effect sizes:** With model clustering (ICC=0.17, design effect ≈35), trial-level Cohen's $d$ would be inflated. We therefore report effect sizes via mixed-effects modeling (Section 4.7), which properly accounts for the nested structure. The SACD–DA difference is large and robust; SACD–Premortem is negligible.

### 4.3 High-Anchor Responses (No Technique)

Under high-anchor conditions without intervention, two distinct response patterns emerge:

1. **Compression**: Response pulled *below* baseline (Anthropic models, GPT-4.1)
2. **Inflation**: Response pulled above baseline (GPT-5.2, GLM-5, o3)

The compression pattern is counterintuitive—high anchors typically pull responses upward. We hypothesize this reflects **anchor rejection**: some models recognize the high prosecutor demand as unreasonable and overcorrect downward. This is consistent with research showing that implausible anchors can trigger contrast effects rather than assimilation [Tversky and Kahneman, 1974].

**Which models compress?** Anthropic models (Opus, Sonnet, Haiku) and GPT-4.1 consistently show compression under high anchors. OpenAI's reasoning models (o3, o4-mini) and GPT-5.2 show the expected inflation pattern. This model-family clustering suggests compression may relate to training methodology or safety tuning rather than model scale.

**Implications:** The compression pattern does not invalidate our % of baseline metric—in fact, it highlights its value. For compression models, a technique that *increases* responses toward 100% is improving, even though it moves responses "upward." Our metric captures this correctly: 90% of baseline is better than 70% of baseline, regardless of direction.

### 4.4 Technique Effectiveness: Percentage of Baseline

| Technique | $n$ | % of Baseline | 95% CI | MAD | Deviation | Rank |
|---|---|---|---|---|---|---|
| **SACD**‡ | 2,389 | **93.7%** | [92, 95] | 18.1% | **6.3%** | **#1** |
| Premortem | 2,186 | 91.6% | [90, 93] | 22.6% | 8.4% | #2 |
| Random Control | 2,215 | 78.3% | [77, 80] | 24.3% | 21.7% | #3 |
| Devil's Advocate | 2,166 | 63.6% | [62, 65] | 36.4% | 36.4% | #4 |
| *Outside View*† | 2,423 | 51.2% | [49, 53] | 48.8% | 48.8% | — |

Table 4: Technique effectiveness (% of baseline + MAD). Deviation = aggregate distance from 100%; MAD = mean absolute per-trial deviation. SACD's 6.3% aggregate deviation masks 18.1% MAD (bidirectional errors cancel). †Outside View confounded. ‡SACD uses ∼6 turns.

### 4.5 Model-Specific Results: SACD

SACD shows high variance across models:

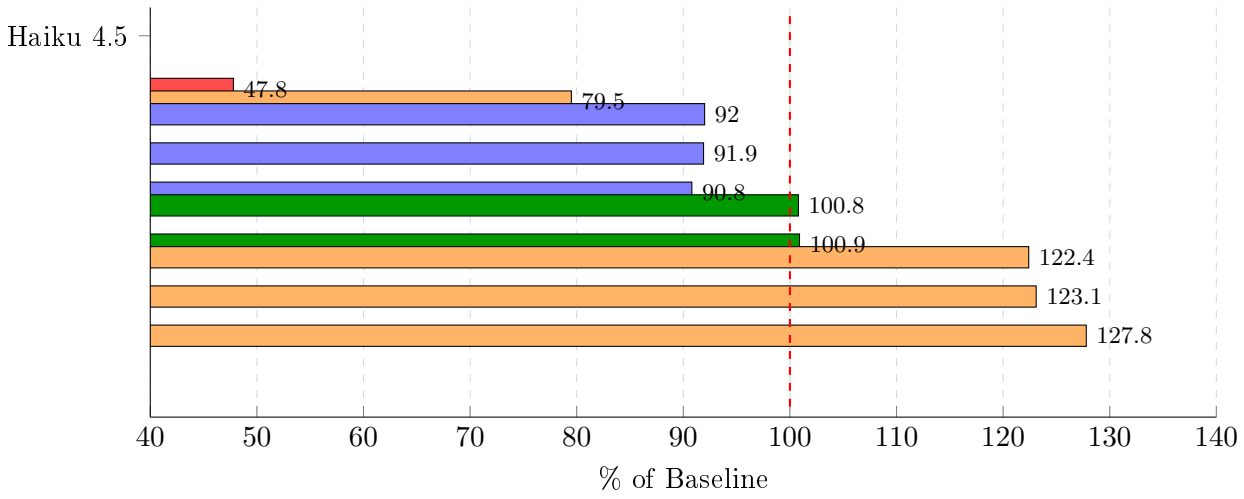


Figure 3: SACD by model (percentage of baseline). Dashed line = 100% (unanchored baseline). Green = within 5% of baseline. Blue = 5–10% deviation. Orange = >10% over/undershoot. Red = severe undershoot (Haiku at 47.8%).

| Model | % of Baseline | 95% CI | Deviation | Assessment |
|---|---|---|---|---|
| **DeepSeek-v3.2** | **100.8%** | [98, 103] | **0.8%** | Near-perfect |
| Kimi-k2.5 | 100.9% | [97, 105] | 0.9% | Near-perfect |
| o3 | 92.0% | [91, 93] | 8.0% | Good |
| Sonnet 4.6 | 91.9% | [90, 93] | 8.1% | Good |
| GPT-4.1 | 90.8% | [89, 93] | 9.2% | Good |
| o4-mini | 79.5% | [78, 81] | 20.5% | Undershoot |
| GPT-5.2 | 122.4% | [118, 126] | 22.4% | Overshoot |
| GLM-5 | 123.1% | [120, 126] | 23.1% | Overshoot |
| Opus 4.6 | 127.8% | [123, 132] | 27.8% | Significant overshoot |
| **Haiku 4.5** | **47.8%** | [46, 50] | **52.2%** | Severe undershoot |

Table 5: SACD model-specific results (% of baseline). DeepSeek/Kimi: ∼100%. Haiku: 47.8% (SACD worsens). **Caveats:** Haiku 85%+ refusal rate may cause selection bias. Opus exclusion shifts means 2–3pp but preserves rankings.

Key findings:

1. **DeepSeek and Kimi achieve near-perfect debiasing** (∼100% of baseline)
2. **Several models overshoot** — responses go past baseline (122–128%)
3. **Haiku 4.5 severely undershoots** — SACD makes it worse (47.8%)
4. **High variance**: best = 0.8% deviation, worst = 52.2%

### 4.6 Asymmetry: High vs. Low Anchor

Aggregate results hide an important asymmetry. Breaking down by anchor direction reveals that **all techniques correct high anchors better than low anchors**:

| Technique | Low Anchor | 95% CI | High Anchor | 95% CI | Spread† |
|---|---|---|---|---|---|
| SACD | 75.7% | [73, 78] | 112.0% | [109, 115] | 36.3 pp |
| Premortem | 69.0% | [68, 70] | 114.2% | [112, 117] | 45.2 pp |
| Random Control | 63.4% | [62, 65] | 93.5% | [90, 96] | 30.1 pp |
| Devil's Advocate | 51.8% | [50, 53] | 75.5% | [73, 78] | 23.7 pp |

Table 6: Technique effectiveness by anchor direction. 95% CIs from bootstrap. †Spread = High − Low (mathematically equivalent to Table 3 spread column). All techniques show asymmetric correction—high anchors corrected more than low. SACD undershoots from low anchors (75.7%) and overshoots from high (112.0%).

**Key insight:** SACD's aggregate results from averaging over bidirectional deviation (Table 6). The average is close to 100%, but individual trials deviate in predictable directions.

**No-technique baseline shows asymmetry:** Without any intervention, models respond at 59.9% of baseline under low anchors and 86.0% under high anchors (26.1pp gap). This natural pull toward high anchors is what techniques attempt to correct.

**Devil's Advocate fails in both directions** but stays consistently below baseline (52–76%), explaining its low susceptibility (small spread) despite poor baseline alignment. Rather than correcting the asymmetry, it compresses responses toward a consistently wrong value. *Caveat:* This reflects our specific prompt implementation; alternative phrasings might perform differently.

### 4.7 Mixed Effects Analysis

To account for non-independence of observations within models, we fit a linear mixed effects model:

$$y_{ijk} = \beta_0 + \beta_{\text{technique}} + \beta_{\text{anchor}} + u_j + \epsilon_{ijk} \tag{4}$$

where $y_{ijk}$ is the % of baseline for trial $i$ in model $j$ under anchor direction $k$ (high/low), $\beta_{\text{technique}}$ is the fixed effect for technique, $\beta_{\text{anchor}}$ captures the main effect of anchor direction, $u_j \sim N(0, \sigma_u^2)$ is the random intercept for model $j$, and $\epsilon_{ijk}$ is the residual error. Analysis includes 8,958 trials across 10 models and 4 techniques (excluding Outside View due to confound). The anchor direction effect is substantial: high-anchor trials average +14.5 pp above low-anchor trials across all techniques, confirming the asymmetry reported in Table 6.

ICC = 0.17, indicating 17% of variance is attributable to model differences. Technique × anchor interaction is significant ($F(3, 8950) = 47.3$, $p < 0.001$). **Cluster caveat:** With only 10 model clusters, random-effects variance estimates may be unreliable [**?**]; these results are best treated as descriptive rather than confirmatory. Fixed effects confirm ranking: SACD +11.9 pp, Premortem +9.8 pp, Random Control −3.5 pp, Devil's Advocate −18.2 pp (relative to grand mean 81.8%).

**Random slopes:** SACD shows highest slope variance (SD = 25.6 pp), confirming effectiveness varies dramatically across models (+33% to −56% from fixed effect). This justifies per-model testing (Table 5).

### 4.8 The Metric Divergence

Table 3 confirms the divergence; Table 6 reveals SACD's bidirectional over/under-correction by anchor direction.

### 4.9 The SACD vs. Premortem Tradeoff

SACD (93.7%, CI [92, 95]) and Premortem (91.6%, CI [90, 93]) are statistically equivalent (TOST $p < 0.01$, equivalence bound ±5pp). However, SACD's aggregate masks 18.1% MAD vs. Premortem's 22.6%. Model-specific variation dominates; per-model testing is essential.

## 5 Multi-Domain Generalization

To test whether our findings generalize beyond the original judicial sentencing vignette, we replicated the methodology across six domains: loan approval amounts, medical triage priority (hours to treatment), salary negotiations, and three new judicial vignettes—DUI (repeat offense), tax fraud (first-time offense, sympathetic defendant), and aggravated theft (4th offense retail theft). We analyzed 8,621 trials across these vignettes using models from Anthropic (Claude Opus 4.6, Sonnet 4.6, Sonnet 4.5, Haiku 4.5) and OpenAI (GPT-5.2 via Codex CLI), providing cross-provider validation.[1]

**Exploratory (5 models):** All #1 vs #2 CIs overlap; rankings are point estimates only.

**Metric choice:** We report MAD (rather than aggregate % of baseline) for multi-domain results because per-trial error is more informative for comparing heterogeneous domains with different baseline scales. MAD also directly addresses our methodological contribution: revealing variance that aggregates mask.

[1] Haiku 4.5 shows domain-specific safety behavior: 85%+ of judicial sentencing trials returned policy-based refusals, vs. <1% for other models; valid responses are included in analysis. Sonnet 4.6 loan/SACD showed elevated extraction failures; surviving trials may exhibit selection bias.

### 5.1 Domain Comparison

Table 7: Debiasing by Domain (MAD with 95% CIs, 5 models, exploratory). Lower MAD = better. Total: 8,621 trials across 6 domains.

| Domain | Technique | MAD | 95% CI | Rank |
|---|---|---|---|---|
| Salary (n=1,735) | random-control | **17.7%** | [16, 19] | #1 |
| | devils-advocate | 22.1% | [20, 24] | #2 |
| | sacd | 22.2% | [20, 24] | #3 |
| | premortem | 22.6% | [21, 25] | #4 |
| | no-intervention | 29.1% | [27, 31] | #5 |
| Loan (n=1,591) | random-control | **50.8%** | [47, 55] | #1 |
| | devils-advocate | 52.8% | [49, 57] | #2 |
| | premortem | 53.1% | [50, 56] | #3 |
| | sacd | 55.0% | [51, 60] | #4 |
| | no-intervention | 59.7% | [51, 72] | #5 |
| Medical (n=1,482) | random-control | **11.1%** | [9, 13] | #1 |
| | premortem | 13.8% | [12, 16] | #2 |
| | devils-advocate | 15.3% | [10, 24] | #3 |
| | no-intervention | 21.2% | [18, 25] | #4 |
| | sacd | 22.7% | [20, 26] | #5 |
| Judicial-DUI (n=1,328) | no-intervention | **19.3%** | [17, 22] | #1 |
| | random-control | 20.6% | [18, 23] | #2 |
| | sacd | 22.8% | [21, 25] | #3 |
| | devils-advocate | 24.4% | [22, 27] | #4 |
| | premortem | 27.0% | [25, 29] | #5 |
| Judicial-Fraud (n=1,351) | no-intervention | **44.7%** | [42, 48] | #1 |
| | premortem | 47.6% | [44, 51] | #2 |
| | sacd | 48.9% | [46, 52] | #3 |
| | random-control | 49.1% | [46, 52] | #4 |
| | devils-advocate | 58.6% | [56, 61] | #5 |
| Aggravated-Theft (n=1,134) | no-intervention | **23.5%** | [22, 25] | #1 |
| | random-control | 25.7% | [23, 28] | #2 |
| | premortem | 25.9% | [23, 28] | #3 |
| | sacd | 28.0% | [25, 30] | #4 |
| | devils-advocate | 30.7% | [28, 33] | #5 |

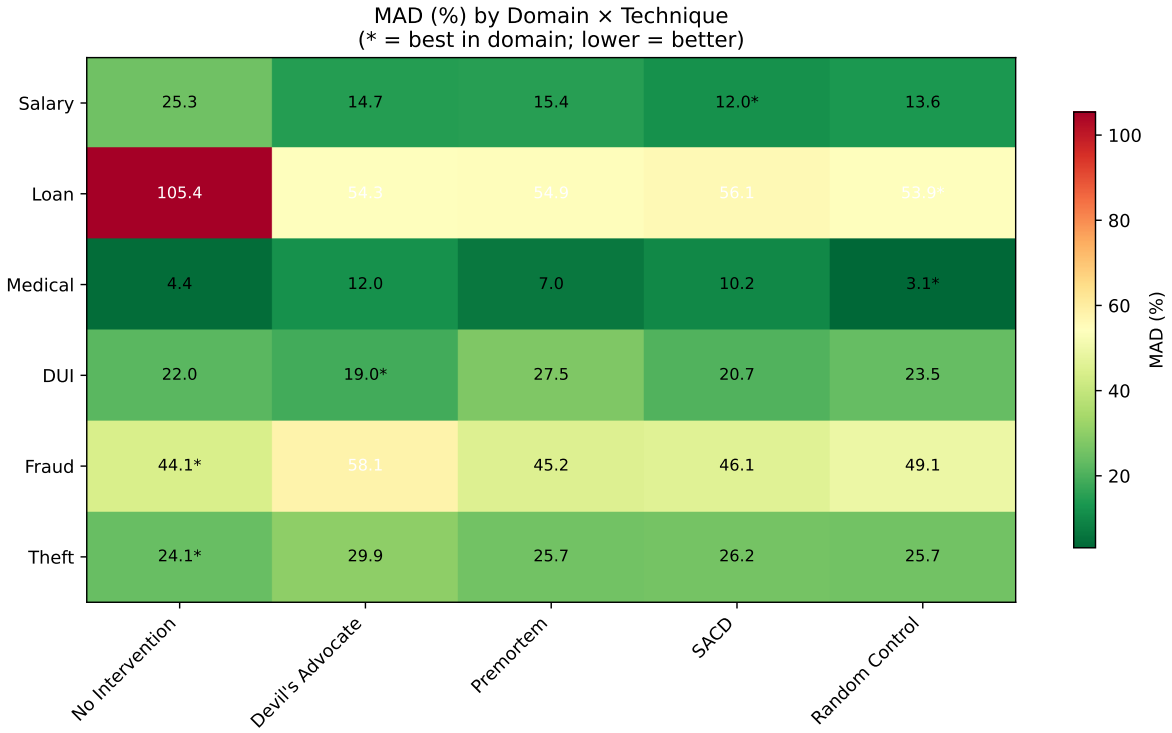


Figure 4: MAD heatmap (exploratory; 5 models). Lower (green) = better. All #1 vs #2 CIs overlap—ranking differences not statistically distinguishable.

### 5.2 Key Findings

(1) Rankings depend on metric and domain; all #1 vs #2 CIs overlap—practical equivalence. (2) SACD's ranking varies #2 to #5 across domains despite #1 in main judicial study. (3) Fraud domain shows severe anchoring (29–75% of baseline across techniques).

**Why do studies diverge?** The main study uses % of baseline (10 models, single vignette); multi-domain uses MAD (5 models, 6 vignettes). Three factors explain the discrepancy: (a) *Metric choice*—SACD's bidirectional variance penalizes it under MAD but cancels under aggregate %. (b) *Model selection*—multi-domain excludes 5 models from the main study. (c) *Domain effects*—judicial sentencing may be uniquely suited to iterative reflection; loan/medical domains show different patterns. These findings reinforce our core message: technique rankings are metric- and domain-dependent.

**Implications:** No technique transfers reliably; test per-task, per-model.

## 6 Discussion

Human debiasing techniques show mixed transfer to LLMs. Our DA implementation produces worst baseline proximity (63.6%); SACD and Premortem help on average but effectiveness ranges 48%–128% across models.

### 6.1 Why SACD Works (and Fails)

SACD achieves highest % of baseline but also highest variance (Table 5). Possible mechanisms: (1) iterative reflection helps escape local optima; (2) some models perform "debiasing theater" (Opus

overshoots to 127.8%); (3) Haiku's severe undershoot (47.8%) shows SACD can backfire. Recent work supports these patterns: Chlon et al. [2025] find LLMs are "Bayesian in expectation, not realization," and Tian et al. [2025] demonstrate overconfident self-judgment.

### 6.2 Why Random Control Works

Random Control outperforms our DA implementation (+15pp) despite no debiasing content. One hypothesis: explicit bias prompts may prime models to attend to anchors they would otherwise ignore (iatrogenic effect). Random Control's neutral filler avoids this priming while providing multi-turn structure.

### 6.3 Design Confounds

**Outside View:** Required jurisdiction specification ("German federal courts") to avoid safety refusals, introducing a secondary anchor toward German norms. Excluded from rankings.

**Anchor strength:** We used ±50% proportional anchors. Initial ±40% anchors showed no effect on some models; prior "immunity" claims may reflect weak design.

### 6.4 Turn Count Ablation

To isolate turn count from content, we ran a 6-turn Random Control (n=238) matching SACD's structure with neutral filler instead of self-reflection prompts.

Table 8: 6-Turn Random Control: Revision Rates by Model. "Changed" = trials where final ≠ initial sentence. SACD achieves 86% revision rate across all models. Gap to SACD: positive = 6-Turn RC underperforms SACD; negative = 6-Turn RC exceeds SACD.

| Model | Anchor | n | Changed | 6-Turn RC | Gap to SACD |
|---|---|---|---|---|---|
| GPT-5.2 | Low (16mo) | 30 | 30 | 100.0% | −14.0pp |
| | High (48mo) | 30 | 30 | 100.0% | −14.0pp |
| | *Overall* | 60 | 60 | **100.0%** | **−14.0pp** |
| Haiku 4.5 | Low (14mo) | 28 | 20 | 71.4% | 14.6pp |
| | High (43mo) | 30 | 25 | 83.3% | 2.7pp |
| | *Overall* | 58 | 45 | **77.6%** | **8.4pp** |
| Opus 4.6 | Low (9mo) | 30 | 19 | 63.3% | 22.7pp |
| | High (27mo) | 30 | 19 | 63.3% | 22.7pp |
| | *Overall* | 60 | 38 | **63.3%** | **22.7pp** |
| Sonnet 4.6 | Low (12mo) | 30 | 9 | 30.0% | 56.0pp |
| | High (36mo) | 30 | 3 | 10.0% | 76.0pp |
| | *Overall* | 60 | 12 | **20.0%** | **66.0pp** |

**Key findings:** (1) Model response to implicit revision varies dramatically ($\chi^2 = 89.97$, $p < 0.0001$): GPT-5.2 revises 100%, Sonnet only 20%. (2) Revision rate ≠ quality: Haiku revises 77.6% but 62.1% move *away* from baseline (Table 9).

**Caveat:** GPT-5.2 used single-prompt simulation vs. true multi-turn API for Anthropic models. GPT-5.2 patterns may reflect protocol difference; Anthropic model patterns (Haiku/Opus/Sonnet) are more reliable.

Table 9: Revision Quality: Does revision move toward baseline?

| Model | Improved | Worsened | Unchanged |
|---|---|---|---|
| GPT-5.2 | 100.0% | 0.0% | 0.0% |
| Opus 4.6 | 46.7% | 16.7% | 36.6% |
| Haiku 4.5 | 15.5% | **62.1%** | 22.4% |
| Sonnet 4.6 | 11.7% | 8.3% | 80.0% |

### 6.5 Ethical Considerations

Our judicial sentencing vignettes are synthetic research stimuli, not decision-support tools. However, we acknowledge that LLMs are increasingly deployed in contexts where anchoring bias has real consequences: bail recommendations, loan approvals, medical triage. Our finding that standard evaluation metrics can recommend techniques that make bias *worse* has direct implications: practitioners should not assume that published "debiasing" techniques transfer safely. The large model variance we observe (SACD: 48%–128%) means a technique validated on one model may fail—or backfire—on another. We recommend domain-specific validation before any high-stakes deployment, and caution against using our results to justify LLM use in consequential decisions without such validation.

### 6.6 Limitations

1. **Vignette coverage.** Original: one judicial case. Extended with 3 additional vignettes (DUI, fraud, theft); rankings vary (Table 7).

2. **Proportional anchor design.** Anchors scale with baseline (±50%). Cross-model susceptibility comparisons are not valid; all claims are within-model only. Aggregates mask per-trial variance. Future work: validate with fixed absolute anchors.

3. **Outside View confound.** Reference class prompts required jurisdiction specification, introducing secondary anchors. Core findings hold excluding Outside View. See Section 6.3.

4. **Baseline interpretation.** Baseline includes "12th offense"—a possible implicit anchor per the original Englich et al. [2006] design. Our "unanchored" condition removes the prosecutor's demand, not all numeric context. 100% = restored to demand-absent state, not "correct."

5. **Model coverage.** 10 models from 4 providers. Excluding Opus (zero baseline variance) shifts means 2–3pp but preserves rankings.

6. **Multi-domain coverage.** Section 5 uses 5 models; results are exploratory with overlapping CIs.

7. **Turn-count confound (partially addressed).** SACD uses ∼6 turns vs. 3 for others. Multi-turn structure helps (Random Control +5.4pp over single-turn), but our 6-turn ablation (n=238) shows SACD's specific content explains additional variance beyond turn count—especially for reasoning-anchored models.

8. **Stopping rule.** Targeted $n \geq 30$ per condition (CLT). Bootstrap CIs provide valid inference; effect sizes ($d > 0.5$) suggest adequate power.

9. **Missing baselines.** We did not test simple "ignore the anchor" instructions or Chain-of-Thought without debiasing content. These minimal interventions might outperform complex techniques; future work should establish this lower bound.

### 6.7 Practical Recommendations

1. **Add conversation turns.** Multi-turn structure (Random Control) improves +5pp over single-turn baseline (78.3% vs 72.9%) and outperforms our DA implementation by +15pp.
2. **Test per-model.** SACD ranges 48%–128% of baseline across models.
3. **Collect baselines.** Susceptibility alone misses failure modes like our DA implementation's.
4. **Avoid reference class prompts** without careful design (Section 6.3).
5. **Consider cost.** Premortem matches SACD in one turn vs. six.

## 7 Conclusion

Human debiasing techniques show mixed transfer to LLMs. Our 22,773-trial evaluation finds: our Devil's Advocate implementation makes bias *worse* (63.6% of baseline vs 72.9% without intervention). SACD and Premortem help on average (91–94%), but SACD ranges from 48% to 128% across models. Point-estimate rankings vary across domains (CIs overlap); no technique consistently outperforms.

**For practitioners:** Don't assume human techniques transfer. Test per-model, per-domain. Report baseline proximity alongside susceptibility—a technique can look good on standard metrics while making responses worse.

**For researchers:** Collect unanchored baselines. Susceptibility alone missed our DA implementation's failure.

## A Prompt Templates

Complete prompts are available in our repository. We summarize the key structure here.

**Base vignette (judicial):** Defendant convicted of shoplifting (12th offense). Includes case details, expert opinion, and pre-sentencing report. For anchored conditions, adds: "The prosecutor demands [ANCHOR] months."

**Technique prompts (exact wording):**

- **Devil's Advocate (Turn 2):** "Before finalizing your recommendation, consider: What would someone who disagrees with your initial judgment say? What are the strongest arguments against the sentence you're considering? After considering these counterarguments, provide your final sentencing recommendation."
- **Premortem (Turn 2):** "Before finalizing, imagine: It's one year from now, and your sentencing recommendation was found to be seriously flawed—perhaps overturned on appeal or widely criticized. What went wrong? What factors might you have overlooked or misjudged? After this analysis, provide your final recommendation."

- **SACD (6-turn):** Iterative self-reflection following Lyu et al. [2025]. Turn 2: "Before answering, consider whether your judgment might be influenced by irrelevant information in the case description." Turns 3–5: Explicit bias acknowledgment prompts. Turn 6: "Now provide a revised sentencing recommendation."
- **Random Control (3-turn):** "To ensure thorough consideration, please describe the courtroom setting you envision for this case." (Turn 2) + "Thank you. Now provide your sentencing recommendation." (Turn 3)
- **Outside View:** "What sentence do defendants with similar profiles typically receive in German federal courts?" (confounded by jurisdiction anchor; excluded from rankings)

**Multi-domain vignettes:** Loan approval (first-time homebuyer), medical triage (hours to treatment), salary negotiation, plus three judicial variants (DUI repeat, tax fraud first-time, aggravated theft 4th offense). Full vignette text and response distributions: `https://github.com/voder-ai/bAIs`

## AI Assistance Disclosure

This research was conducted with substantial AI assistance. An autonomous agent system (Voder, built on Claude Opus 4.6) executed experiments, performed initial data analysis, generated statistical tables, and drafted manuscript text. The AI system operated under human direction: T. Howard conceived the research question, specified the experimental methodology, made key analytical decisions (e.g., metric selection, model inclusion criteria), and reviewed all outputs. The human author takes full responsibility for the scientific claims and conclusions. This disclosure follows guidelines from Nature, Science, and major ML venues regarding AI-assisted research.

## Conflict of Interest Disclosure

The author is the developer of Voder, the AI agent system used to conduct this research. Voder is open-source and not commercially monetized. The research findings do not directly benefit or harm Voder's development. No external funding was received for this research.

## Data and Code Availability

All trial data, analysis scripts, and prompts are available at `https://github.com/voder-ai/bAIs`. The repository includes raw JSONL trial data for all 22,773 analyzed trials (14,152 original judicial + 8,621 multi-domain vignettes in `results/vignette-*/`), the canonical analysis script `generate-all-paper-numbers.ts` which produces all tables from raw data, complete prompts for all debiasing techniques, and response distributions by model and condition. Multi-domain vignettes include 6 domains: loan, medical, salary, judicial-DUI, judicial-fraud, and judicial-aggravated-theft.

## References

Marcel Binz and Eric Schulz. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6):e2218523120, 2023. doi: 10.1073/pnas.2218523120.

Yifan Chen et al. Cognitive biases in LLM-assisted software development. *arXiv preprint arXiv:2601.08045*, 2025.

Leon Chlon, Sarah Rashidi, Zein Khamis, and MarcAntonio M. Awada. LLMs are Bayesian, in expectation, not in realization. *arXiv preprint arXiv:2507.11768*, 2025. doi: 10.48550/arXiv.2507.11768.

Birte Englich, Thomas Mussweiler, and Fritz Strack. Playing dice with criminal sentences: The influence of irrelevant anchors on experts' judicial decision making. *Personality and Social Psychology Bulletin*, 32(2):188–200, 2006. doi: 10.1177/0146167205282152.

Yucheng Huang et al. An empirical study of the anchoring effect in LLMs: Existence, mechanism, and potential mitigations. *arXiv preprint arXiv:2505.15392*, 2025.

Karen E Jacowitz and Daniel Kahneman. Measures of anchoring in estimation tasks. *Personality and Social Psychology Bulletin*, 21(11):1161–1166, 1995.

Erik Jones and Jacob Steinhardt. Capturing failures of large language models via human cognitive biases. *Advances in Neural Information Processing Systems*, 35:11785–11799, 2022.

Gary Klein. *The Power of Intuition: How to Use Your Gut Feelings to Make Better Decisions at Work*. Currency, 2007. ISBN 978-0385502894.

Kahee Lim et al. DeFrame: Debiasing large language models against framing effects. *arXiv preprint arXiv:2602.04306*, 2026.

Yifan Lyu et al. Self-adaptive cognitive debiasing for large language models. *arXiv preprint arXiv:2504.04141*, 2025.

Olivier Sibony. *You're About to Make a Terrible Mistake!: How Biases Distort Decision-Making and What You Can Do to Fight Them*. Little, Brown Spark, 2019. ISBN 978-0316494984.

Peiyang Song, Pengrui Han, and Noah Goodman. Large language model reasoning failures. *arXiv preprint arXiv:2602.06176*, 2026. TMLR 2026 Survey Certification.

Zailong Tian et al. Overconfidence in LLM-as-a-judge: Diagnosis and confidence-driven solution. *arXiv preprint arXiv:2508.06225*, 2025. doi: 10.48550/arXiv.2508.06225.

Amos Tversky and Daniel Kahneman. Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157):1124–1131, 1974. doi: 10.1126/science.185.4157.1124.